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SUTD DESIGN•AI CHALLENGE

SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN

[Energy Saving Mode] Group Proposal

We declare that all ideas submitted are humanly made and only visuals and materials we chose are created with AI assistance.

School / Institution:

Team Member:

S/No.	Name	Academic Level
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PROBLEM STATEMENT

Energy security has always been an important aspect for Singapore and even more so in recent years. As a small country with limited natural resources, how might we enhance national energy security so that Singaporeans can enjoy affordable and sustainable electricity for the long term?

MATERIAL REQUIRED (FOR MOCK UP ONLY)

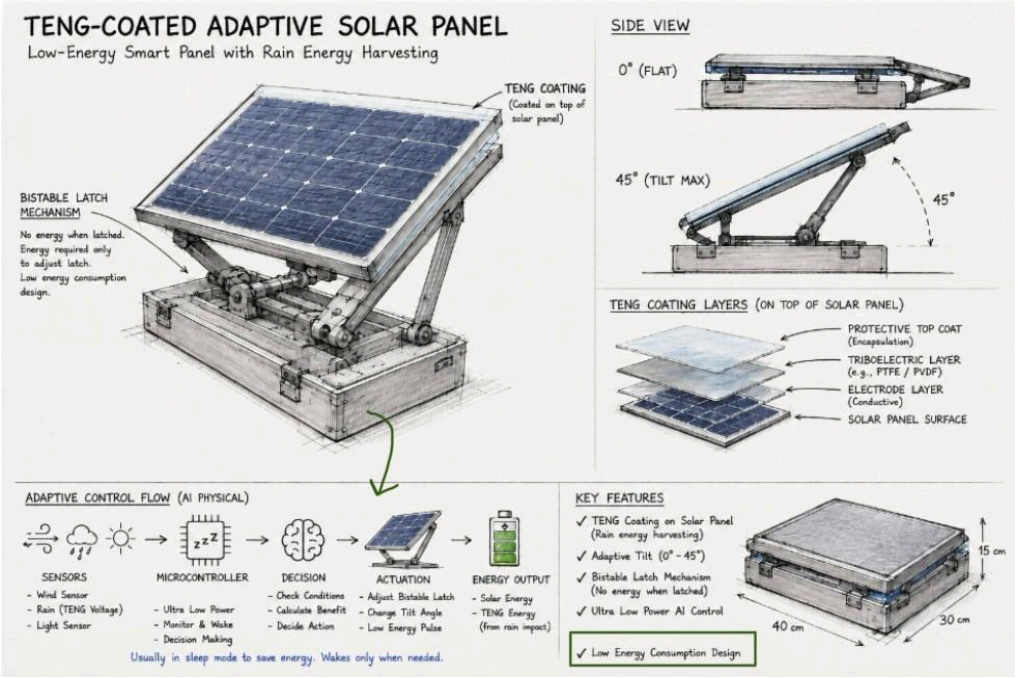
Materials	Quantity	Remarks
silica gel	10 packets	to mimic actual desiccant which is expensive probably and we need experiments
bottle of water or just water	1 bottle can refill	to mimic humid air
acrylic sheet/foam board/ cardboard	3 large sheets	to build structure/base make a box for the stuff
battery	some	to power set up
breadboard	small 2 pieces	for circuit to be built
wires	many	for circuit to be built

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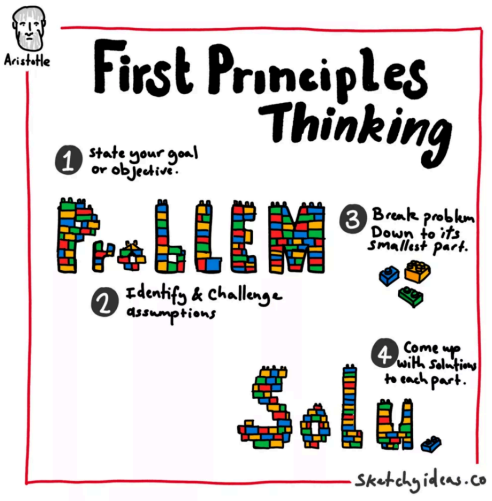
motor that is able to spin a tray with 10 silica gel packets small	1 unit	rotates the dual-bed tray 180° when triggered
heating element	1 unit	to act as hot refrigerant
battery holders	some	to hold battery
microcontroller/ the programmed thingy that does stuff and triggers the motor to on	1 unit	we can pour water on silica gel and use the fact that the dry gel insulator has water fill the gaps and make a complete circuit

We do not have any engineering knowledge on circuitry, mechanical components and more to accurately decide on the materials required for our physical AI build. As such our materials needed are not very in depth and we appreciate assistance…

IDEATION (two to three ideation)

Ideation 1 (Description and/ or sketch)	Our Solution: AERIS-01 (Adaptive Energy & Rain Integrated Solar-01) Design Sketch By ChatGPT:  AERIS-01 is a smart solar panel set up designed to optimize energy yield and offers dual energy collection from sunlight and rain conditions. The solar panel is layered with TENG technology which makes use of friction between water particles in rain and the panel to generate electricity in rain alongside the photovoltaic cells that generate energy from sunlight. AERIS-01 also maximizes net energy gain by calculating whether movement and tilt to optimize energy yield uses more energy than resulting energy yield

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	increase and executes movement accordingly to ensure maximum energy yield increase. Furthermore, the microcontroller responsible for the logic and decision making of the system remains mainly in a deep-sleep mode and only awakens during sudden environmental change, ensuring the system uses the least amount of energy possible. AERIS-01 is the future of Singapore's renewable energy, making unpredictable weather into a sustainable and reliable energy source and thus contributing to the strengthening of Singapore's Energy Security. Refer to the Algorithm/Mode of Employment of AERIS-01 under the ideation segment.
Ideation Process and Elaboration	The main problem -> Lack of energy security [FUTURE] ❖ Current fossil fuels powering 95% of our energy needs are not sustainable in the long run so we investigated alternative energy sources that were more sustainable ❖ Considering the lack of geothermal energy sources, and lack of land and sufficient constant stream of wind or water flow for hydropower or windmill powered energy sources, we investigated the only thing that felt feasible enough to be worked on: Solar Power ❖ Limitations of solar energy Cloudy weather, rain and night conditions prevented energy collection Shifting angle of light rays from the sun throughout the day results in inefficient energy storage/gain ❖ We therefore used First Principles Thinking to engineer our solution: 

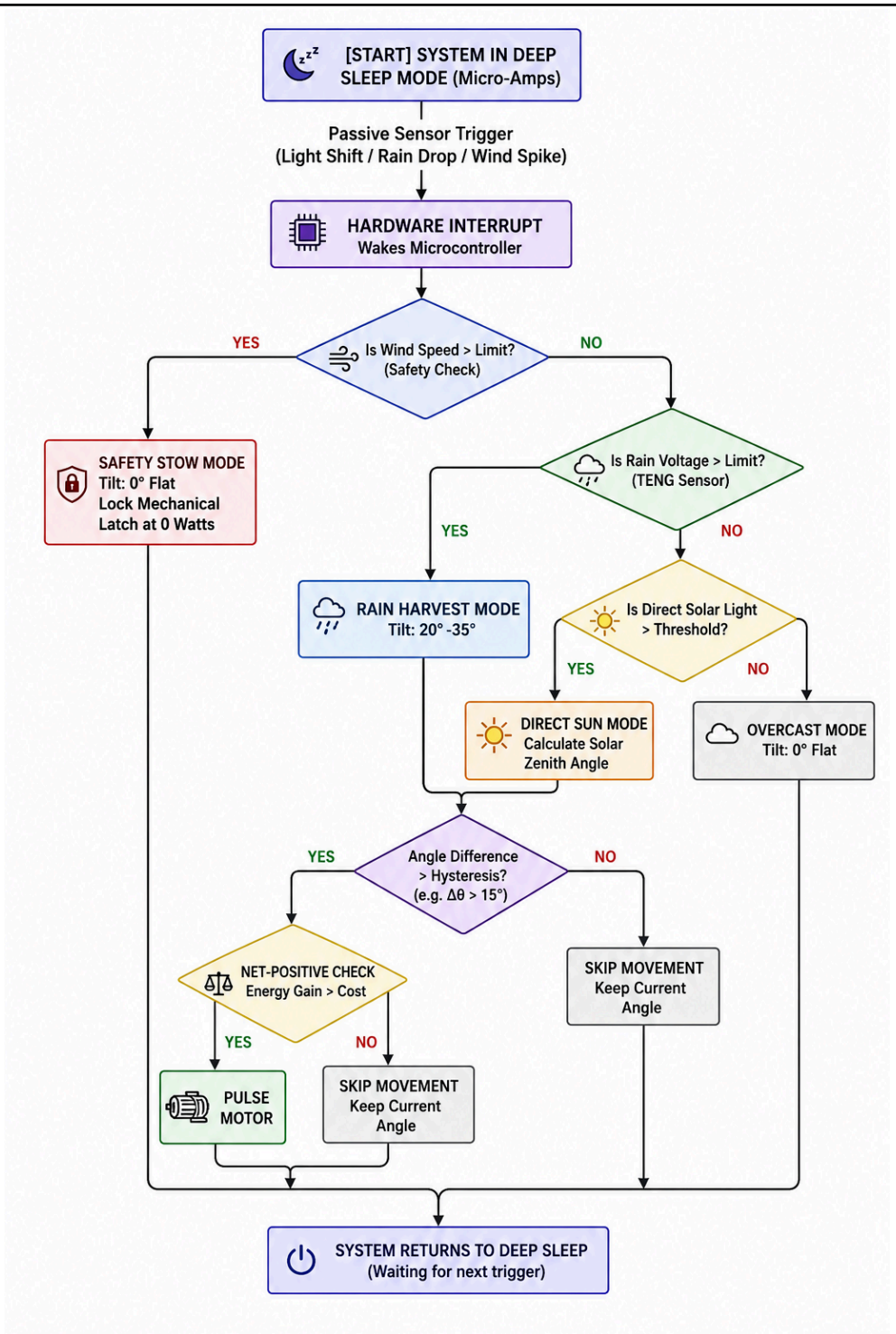
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	❖ First Stating Goal/Objective: Enhance Singapore's renewable energy yield during variable weather conditions ❖ Next Identifying Challenges and Assumptions For maximum solar energy yield: sunny and cool weather Angle between sunlight rays and solar panel to be as close as possible to 90 degrees for maximum energy yield Rain decreases yield of energy captured by solar panels to about 25% of maximum capacity Rain can be converted into energy using Triboelectric Nanogenerator or TENG technology The most efficient angle for a TENG panel to be at is 45 degrees tilt to ensure that most energy is harvested by the system Physical AI systems require energy and result in parasitic draw which reduces maximum or results in no energy yield – using a system that conserves energy with a deep sleep mode and is only activated when significant environmental difference occurs can help reduce energy usage, reduce parasitic draw and result in maximum yield ❖ The third step, breaking the problem down into smallest parts possible How to maximize the yield of energy in sunny weather as time changes (angle of suns light rays changes) How to harvest more energy in rain How to maximize energy yield in rain How to prevent excessive energy use to achieve the above ❖ Finally Addressing each part of the problem Implement dynamic, heliotropic tilt mechanisms that adjust panel angles to ensure sunlight rays strike as close to 90° (perpendicular) as possible throughout the day. Switch to a flat 0° orientation during overcast or diffuse light conditions to capture 360° scattered ambient light.

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	• Integrate a transparent Triboelectric Nanogenerator (R-TENG) surface layer over the photovoltaic cells to convert the kinetic impact energy of falling raindrops into electricity. • Adjust the physical panel inclination to an optimal runoff angle ($\approx 30^{\circ} - 45^{\circ}$) during downpours to balance raindrop impact force with fast water shedding. • Utilize an event-driven hardware architecture where the system remains in a micro-amp deep sleep state. The system wakes up only when environmental triggers (light shifts or raindrop impacts) trip an interrupt, executing a short pulse to lock a bistable mechanical latch that holds the angle at $0\ W$ power draw. Algorithm for our Physical AI

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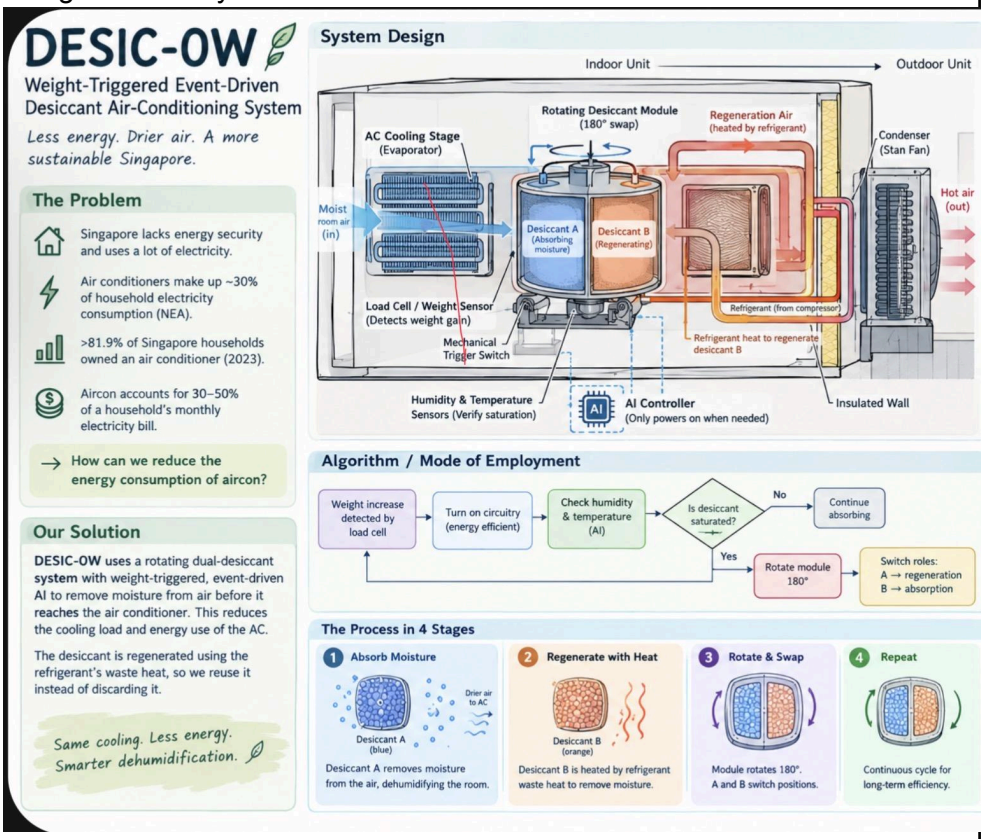


Limitations

- Current TENG technology harvest very little energy from rain and this can mostly be used to power maybe the microcontrollers maybe provide light or power a possible wiping mechanism to clean the solar panel in the future after being stored as energy

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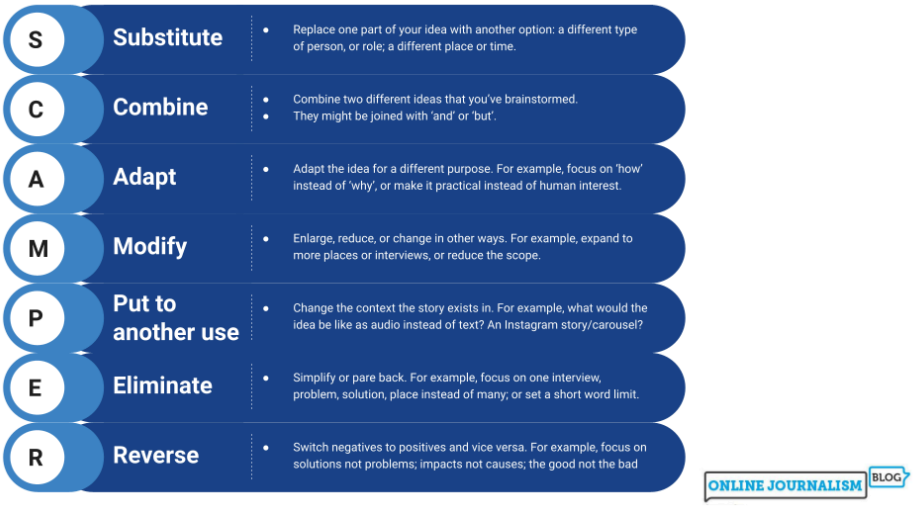
	- Much evolution needed before the TENG technology can give sufficient energy from rain

Ideation 2 (Description and/ or sketch)	Our solution: DESIC-0W (Weight-Triggered Event-Driven Desiccant Air-Conditioning System) Design Sketch By ChatGPT:  DESIC-0W is a desiccant dehumidification air conditioner system with the aim of reducing the energy usage of air conditioners. The system makes use of 2 desiccants. One of them will be used to remove moisture from the air to dehumidify the room in which the air conditioner is. The other will be a desiccant that was previously used and will be heated to remove moisture so it can be reused. . This helps to reduce the energy usage of the air conditioner as rather than using energy to dehumidify the room, chemical processes are being used instead. Furthermore the heat used to regenerate the desiccant to be used again is also from existing air conditioning processes using heat generated after the air conditioner's refrigerant passes through the compressor. Desiccant A -> absorbs water, Desiccant B -> gets regenerated. A and B will be placed on either end of a rotating device. As A reaches its saturation point, it will gain weight from moisture absorption and trigger a

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	mechanical switch which will trigger our AI to turn on. It will then check and determine whether the weight gain by the desiccant was caused by other factors such as physical parts breaking down in air conditioning and falling on the desiccant or was caused by the moisture absorption using humidity sensors to help check saturation moisture absorption. Our humidity sensor will monitor the air around the desiccant to determine the humidity and whether the desiccant is at its maximum moisture holding capacity. Then, if the AI determines that the desiccant is at its maximum moisture holding capacity, the rotating device containing A and B will rotate 180° and then the dry desiccant B will start moisture absorption and A will get regenerated. DESIC-0W also minimises energy usage by using a system where the circuitry remains turned off and only turns on as the weight of the desiccant reaches its maximum capacity. DESIC-0W will help to conserve energy in Singapore by reducing energy usage from the usage of air conditioners. Air conditioners are widely used by many households in Singapore and require a lot of energy to operate. This will help to save a lot of energy and electricity in the long term as more sustainable alternatives are being developed. Refer to the Algorithm/Mode of Employment of DESIC-0W under the ideation segment.
Ideation Process and Elaboration	The main problem -> Singapore lacks energy security and uses a lot of electricity -> 30% of electricity consumption in a household is Aircon -> How can we reduce energy consumption of AC ❖ Air conditioners make up about ~30% of energy consumption in a typical household (NEA website) and over 81.9% of Singapore resident households owned an air conditioner in 2023. ❖ Air conditioners account for 30%-50% of a household monthly electricity bill for many Singaporeans. ❖ We used the SCAMPER method to create our solution to reduce the energy and electricity consumption of air conditioners.

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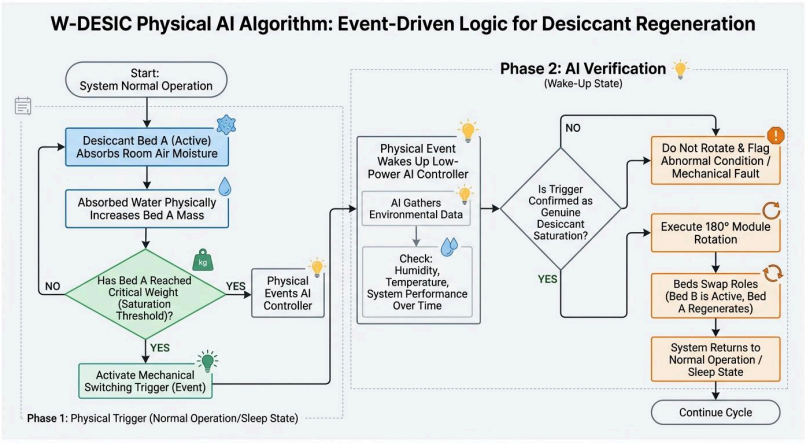
- Put to another use:
  - While researching the energy consumption of air conditioners, we found out that air conditioners waste a lot of energy, by removing moisture from the humid air, to the latent heat of condensation of water whilst cooling the air. We then thought about how many people put silica gel packets in their cupboards to reduce moisture. Hence, we thought that we could use a similar chemical method to reduce moisture of humid air before the air conditioner deals with it, thus reducing the energy used by the air conditioner, as now the air conditioner will only have to cool the dry air.
- Adapt:
  - While silica gel is used in our everyday lives to remove moisture on a small scale, we realised that we could not use regular silica gel packets to remove moisture from a whole room. Hence, we thought of how we could adapt our idea to remove moisture from a room with something stronger. Upon researching, we found out that silica gel is a desiccant. A desiccant is a substance that removes moisture from the surrounding air. We thought that we could use a stronger desiccant to remove moisture from a room. An example of a desiccant which can do so is calcium chloride. However to actually determine which specific desiccant we pull use in our air conditioning we would need to conduct experiments and test which desiccant is most suitable in terms of rate of moisture absorption and retention and how quickly and at what temperature the desiccant could be regenerated to absorb water upon saturation

Speaking of regenerating a desiccant to further absorb moisture, we realised the desiccant will reach its maximum moisture holding capacity and the moisture will need to be dried off. At first, we thought of manually removing the desiccant and drying it off. But how would one know that the desiccant has reached its capacity? We also thought about how when the

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	desiccant is drying, there won't be a dessicant in the air conditioner anymore to remove moisture. Also the possible energy wastage in drying desiccant would mean we won't actually reduce energy consumption of desiccants. ❖ Substitute: We thought of substituting the moist desiccant with a dry desiccant while the moist one was being regenerated. This helps to ensure continuous moisture absorption from the room. As we were researching, we realised we could evaporate the water from the desiccant using the hot refrigerant in the air conditioners refrigeration cycle. Usually, that heat is ejected outside. ❖ Combine: Combining both ideas, we came up with the idea of swapping the two desiccants around once one becomes moist. Then, cooling the moist one with the heat from the refrigerant We thought of how we could use physical AI to modify our idea. Currently, we would have to manually switch the two desiccants around and we would not be sure of when the desiccant has reached its full capacity. At first we thought of having sensors to determine the humidity and weight of the desiccant. The AI would determine whether the weight gain was only by the increase in saturation of the desiccant or by any external factors. However, we would still have to manually switch the desiccants. ❖ Modify: We thought of modifying our existing set up by having rotating plates. The AI would decide when to switch the plates around and there will be one desiccant on each plate. Thus, once the desiccant has reached its maximum moisture holding capacity, the AI will cause the plate to rotate so that we would not have to manually switch the desiccants around anymore. We also wanted to ensure the sensors and AI would not use too much energy so we turned off the circuitry for it and created a model where the weight gain of a desiccant absorbing moisture physically turns on circuitry for AI to check humidity at a certain weight threshold. Algorithm for our Physical AI:

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Limitations

- It might be difficult to install this new system in air conditioners as the process of installing this will be different depending on the air conditioner. This introduces adoption problems.

PROPOSED SOLUTION

Chosen Idea	DESIC-0W
Justification	

Idea	Real	Win	Worth
AERIS-01	This idea is technically feasible, all technology used already exists. This idea increases the energy yield of solar panels which the government may be interested in as it can help achieve the sustainable development goal of affordable and clean energy. However, there would be high upfront costs to develop and sell this product especially	Our product does improve energy yield and thus has an advantage against existing solar panels. While TENG coated solar panels do exist our unique features of moving our panels to maximise energy yield makes it more competitive.	This idea will fit Singapore's long term goals of adopting clean, renewable energy. Furthermore, this will be profitable in Singapore in the long term as it will cost less to obtain energy through this idea. Technological advancements

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		since TENG technology is relatively new. Furthermore the increased energy yield is minimal and as such until TENG technology or other alternatives improve it is not very feasible to work in the market as energy yield from rain is minimal.	(We are going to ignore the company or team skills as it is not relevant for the competition… but I do believe that other professionals may have an advantage over our team of 2 people since we have no engineering related or coding or anything experience)	that make more significant energy gain from rain can be incorporated into our product in the future to help us remain competitive.
	DESIC-0W	It is technically feasible to build with minor adjustments if needed. It helps to decrease energy usage in a home so consumers will be willing to purchase it. Under the enhanced climate friendly households programme the government gives vouchers to households to purchase energy efficient products like our aircon solution. Considering that almost all households in SG have aircon and need air conditioning as well as offices and buildings, there is an	Our product saves energy and reduces people's costs in the long term in their electricity build. Vouchers subsidising the product also makes it more economical and people are increasingly being aware of sustainability in today's world and would thus choose our product over others in the market. It is also a unique solution with less	This will be profitable for households in the long term as they will save money on their energy costs. Air conditioners are one of the highest energy consuming appliances in a house and saving energy on them will reduce the energy bills significantly. This will reduce energy usage in households which lead to a lot of energy being saved in the long term.

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		existing market. Singapore wants to have 80% of buildings attain green certification by 2030, using our air conditioning hugely reduces energy needs creating demand beyond households.	competition as typically only industrial large ai Econs have desiccant systems and we are introducing this to a smaller scale to households and buildings . We may however lack the technical skills to produce and sell this over professionals in the industry	

DESIC-0W has a wider impact than our AERIS-01 as the improvement in energy yield is not as high as how much energy could be saved by changing all air conditioning to DESIC-0W as air conditioners are everywhere and energy yield of TENG technology is lower. We also believe that DESIC-0W may have greater adoption by people due to government subsidies and goals.